

2015-2016 Fall
MAT123 Final
12/01/2016

1. Find the area of the region bounded by the circle $x^2 + y^2 = 1$ and the graph of the absolute value function $y = |x|$.
2. Use the Washer Method to find the volume of the solid obtained by revolving the region bounded by the parabola $y = 2x^2 - 3$ and the curves $y = -3$, $x = 2$ about the line $y = 7$.
3. Use the Cylindrical Shell Method to find the volume of the solid obtained by revolving the region bounded by $y = x^2$ and $y = -x + 1$ about the line $x = -1$.
4. Evaluate the following integrals.

a. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

b. $\int \frac{x^5 + x^4 - 8x^3 + 10x^2 + 12x}{x^2 - 3x + 2} dx$

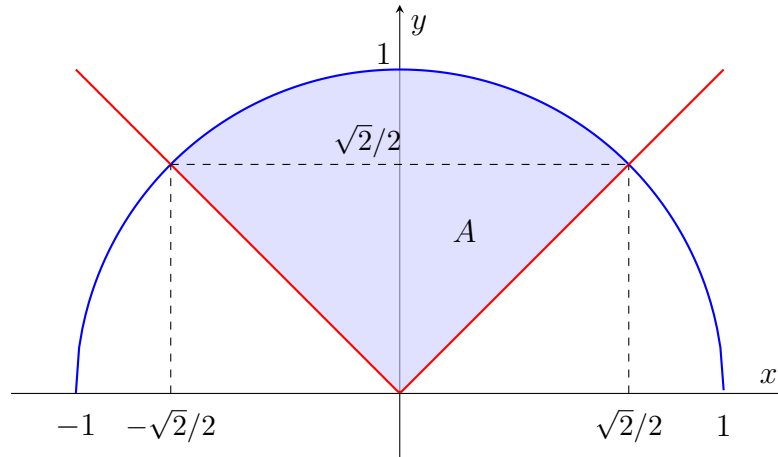
c. $\int \arccos x dx$

d. $\int \frac{1}{x^2 + 3x + 1} dx$

e. $\int \frac{\sin x}{1 + \sin x} dx$

5. Find the surface area of the revolution by rotating the curve $y = e^x$, $0 \leq x \leq 1$ about the x -axis.
6. Investigate the convergence of the improper integral $\int_0^{-\infty} \frac{e^{-x}}{1 + e^{-x}} dx$.
7. Using the Monotone Convergence Theorem, investigate the convergence of the sequence $\left(\frac{\ln n}{n}\right)_{n \in \mathbb{N}}$.

1.



$$\begin{aligned}
 A &= \int_{-\sqrt{2}/2}^{\sqrt{2}/2} \left(\sqrt{1-x^2} - |x| \right) dx \\
 &= \int_{-\sqrt{2}/2}^0 \left(\sqrt{1-x^2} + x \right) dx + \int_0^{\sqrt{2}/2} \left(\sqrt{1-x^2} - x \right) dx \\
 &= \int_{-\sqrt{2}/2}^{\sqrt{2}/2} \sqrt{1-x^2} dx - \int_0^{\sqrt{2}/2} 2x dx \tag{1}
 \end{aligned}$$

Calculate the right-hand integral (??).

$$\int_0^{\sqrt{2}/2} 2x dx = x^2 \Big|_0^{\sqrt{2}/2} = \left(\frac{\sqrt{2}}{2} \right)^2 - (0)^2 = \frac{1}{2}$$

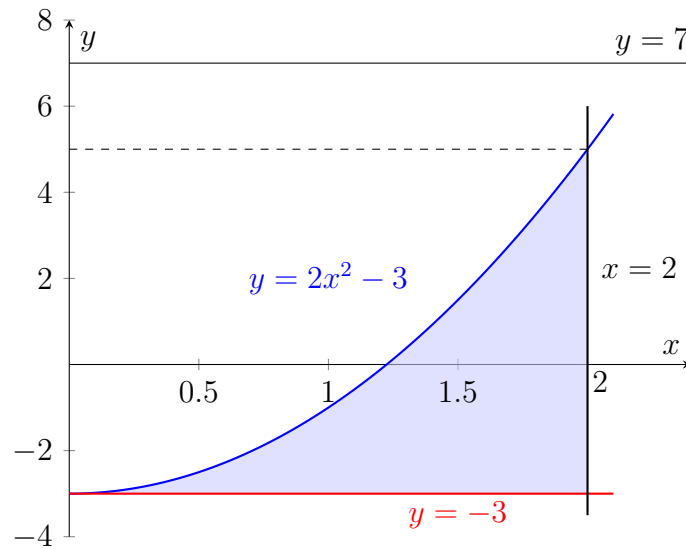
To calculate the left-hand integral in (??), we will use a trigonometric substitution. Let $x = \sin u$, then $dx = \cos u du$.

$$x = -\frac{\sqrt{2}}{2} \implies u = \arcsin \left(-\frac{\sqrt{2}}{2} \right) = -\frac{\pi}{4}, \quad x = \frac{\sqrt{2}}{2} \implies u = \arcsin \left(\frac{\sqrt{2}}{2} \right) = \frac{\pi}{4}$$

$$\begin{aligned}
 \int_{-\sqrt{2}/2}^{\sqrt{2}/2} \sqrt{1-x^2} dx &= \int_{-\pi/4}^{\pi/4} \sqrt{1-\sin^2 u} \cos u du = \int_{-\pi/4}^{\pi/4} |\cos u| \cos u du \quad [|\cos u| > 0] \\
 &= \int_{-\pi/4}^{\pi/4} \cos^2 u du = \int_{-\pi/4}^{\pi/4} \frac{1 + \cos 2u}{2} du = \left[\frac{u}{2} + \frac{\sin 2u}{4} \right]_{-\pi/4}^{\pi/4} \\
 &= \left(\frac{\pi}{8} + \frac{1}{4} \right) - \left(-\frac{\pi}{8} - \frac{1}{4} \right) = \frac{\pi}{4} + \frac{1}{2}
 \end{aligned}$$

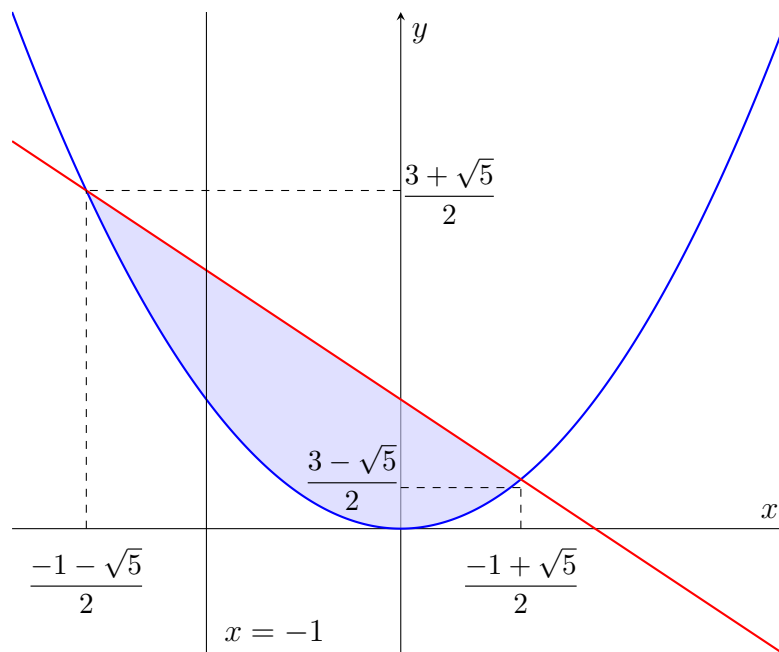
Evaluate (??). The area is $A = \left(\frac{\pi}{4} + \frac{1}{2} \right) - \left(\frac{1}{2} \right) = \boxed{\frac{\pi}{4}}$.

2.



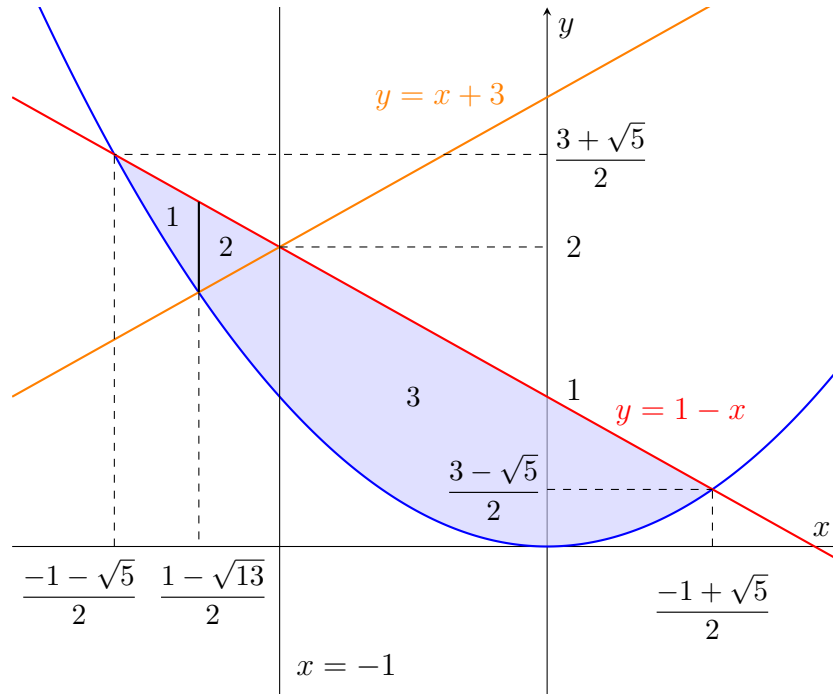
$$\begin{aligned}
 V &= \int_D \pi [R_2^2(x) - R_1^2(x)] \, dx = \int_0^2 \pi [(7 - (-3))^2 - (7 - (2x^2 - 3))^2] \, dx \\
 &= \pi \int_0^2 [(10)^2 - (10 - 2x^2)^2] \, dx = \pi \int_0^2 (100 - 100 + 40x^2 - 4x^4) \, dx \\
 &= \pi \int_0^2 (40x^2 - 4x^4) \, dx = \pi \left[\frac{40x^3}{3} - \frac{4x^5}{5} \right]_0^2 = \pi \left[\left(\frac{320}{3} - \frac{128}{5} \right) - (0) \right] = \boxed{\frac{1216\pi}{15}}
 \end{aligned}$$

3.



Notice that the rotation axis passes through the region. Consider the right-hand region. If we rotate it around the axis, a piece of the region on the left will be inside the revolution. That is, the region bounded by the line that is symmetric to the line

$y = -x + 1$ around $x = -1$ and the curve $y = x^2$. We do not need to rotate that region since it would lead to double revolution. The upper part of the left-hand region may be rotated around $x = -1$ independent of the right-hand region. Divide the left-hand region into three subregions. We get three different subregions to integrate over.



Let V be the volume of the solid.

$$\begin{aligned}
 V &= \int_D 2\pi \cdot r(x) \cdot h(x) dx \\
 &= \int_{\frac{-1-\sqrt{5}}{2}}^{\frac{1-\sqrt{13}}{2}} 2\pi(-1-x) [(1-x) - x^2] dx + \int_{\frac{1-\sqrt{13}}{2}}^{-1} 2\pi(-1-x) [(1-x) - (x+3)] dx \\
 &\quad + \int_{-1}^{\frac{-1+\sqrt{5}}{2}} 2\pi(x+1) [(1-x) - x^2] dx \\
 &= 2\pi \int_{\frac{-1-\sqrt{5}}{2}}^{\frac{1-\sqrt{13}}{2}} (-1 + 2x^2 + x^3) dx + 2\pi \int_{\frac{1-\sqrt{13}}{2}}^{-1} (2x^2 + 4x + 2) dx \\
 &\quad + 2\pi \int_{-1}^{\frac{-1+\sqrt{5}}{2}} (1 - 2x^2 - x^3) dx \\
 &= 2\pi \left\{ \left[-x + \frac{2x^3}{3} + \frac{x^4}{4} \right]_{\frac{-1-\sqrt{5}}{2}}^{\frac{1-\sqrt{13}}{2}} + \left[\frac{2x^3}{3} + 2x^2 + 2x \right]_{\frac{1-\sqrt{13}}{2}}^{-1} + \left[x - \frac{2x^3}{3} - \frac{x^4}{4} \right]_{-1}^{\frac{-1+\sqrt{5}}{2}} \right\}
 \end{aligned}$$

$$\begin{aligned}
V = 2\pi & \left[\frac{\sqrt{13}-1}{2} + \frac{(1-\sqrt{13})^3}{12} + \frac{(1-\sqrt{13})^4}{64} - \frac{1+\sqrt{5}}{2} + \frac{(1+\sqrt{5})^3}{12} - \frac{(1+\sqrt{5})^4}{64} \right] \\
& + 2\pi \left[-\frac{2}{3} + 2 - 2 - \left(\frac{(1-\sqrt{13})^3}{12} + \frac{(1-\sqrt{13})^2}{2} + 1 - \sqrt{13} \right) \right] \\
& + 2\pi \left[\frac{-1+\sqrt{5}}{2} - \frac{(-1+\sqrt{5})^3}{12} - \frac{(-1+\sqrt{5})^4}{64} - \left(-1 + \frac{2}{3} - \frac{1}{4} \right) \right]
\end{aligned}$$

After some mathematical operations, the volume is found to be

$$V = \frac{13\pi}{6} - \frac{\pi}{4} (47 - 13\sqrt{13}) = \boxed{\frac{\pi (39\sqrt{13} - 115)}{12}}$$

The volume can also be evaluated by integrating over the domain and subtracting the region that is revolved twice. That is, the region beneath the region 2 mapped on the graph.

4. a. Let $u = \sqrt{x}$, then $du = \frac{1}{2\sqrt{x}} dx$.

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int \sin u \cdot 2 du = -2 \cos u + C = \boxed{-2 \cos \sqrt{x} + C, \quad C \in \mathbb{R}}$$

- b. Perform a long polynomial division and rewrite the integral in two expressions.

$$\begin{aligned}
I &= \int \frac{x^5 + x^4 - 8x^3 + 10x^2 + 12x}{x^2 - 3x + 2} dx = \int \left(x^3 - 4x^2 - 22x - 48 + \frac{-88x + 96}{x^2 - 3x + 2} \right) dx \\
&= \int (x^3 + 4x^2 + 2x + 8) dx + \int \frac{32x - 16}{(x-2)(x-1)} dx \tag{2}
\end{aligned}$$

Calculate the integral on the left in (??).

$$\int (x^3 + 4x^2 + 2x + 8) dx = \frac{x^4}{4} + \frac{4x^3}{3} + x^2 + 8x + C_1$$

Calculate the integral on the right in (??). Decompose the expression into partial fractions.

$$\begin{aligned}
\int \frac{32x - 16}{(x-2)(x-1)} dx &= \int \left(\frac{A}{x-2} + \frac{B}{x-1} \right) dx \\
32x - 16 &= A(x-1) + B(x-2) = x(A+B) - A - 2B
\end{aligned}$$

Equate the coefficients of like terms.

$$\left. \begin{array}{l} A + B = 32 \\ -A - 2B = -16 \end{array} \right\} \rightarrow A = 48, \quad B = -16$$

Substitute the values into A and B .

$$\int \left(\frac{A}{x-2} + \frac{B}{x-1} \right) dx = \int \left(\frac{48}{x-2} - \frac{16}{x-1} \right) dx = 48 \ln |x-2| - 16 \ln |x-1| + C_2$$

Rewrite (??).

$$I = \frac{x^4}{4} + \frac{4x^3}{3} + x^2 + 8x + 48 \ln |x-2| - 16 \ln |x-1| + C, \quad C \in \mathbb{R}$$

c. Apply integration by parts.

$$\left. \begin{array}{l} u = \arccos x \rightarrow du = -\frac{1}{\sqrt{1-x^2}} dx \\ dv = dx \rightarrow v = x \end{array} \right\} \rightarrow \int v du = uv - \int v du$$

$$\int \arccos x dx = x \arccos x - \int \frac{-x}{\sqrt{1-x^2}} dx$$

Let us use a u -substitution for the integral on the right. Let $u = 1 - x^2$, then $du = -2x dx$

$$\int \frac{-x}{\sqrt{1-x^2}} dx = \int \frac{du}{2\sqrt{u}} = \sqrt{u} + C = \sqrt{1-x^2} + C$$

Therefore,

$$\int \arccos x dx = \boxed{x \arccos x - \sqrt{1-x^2} + C, \quad C \in \mathbb{R}}$$

d. Use the method of partial fraction decomposition.

$$\begin{aligned} I &= \int \frac{dx}{x^2 + 3x + 1} = \int \frac{dx}{x^2 + 3x + \frac{9}{4} - \frac{5}{4}} = \int \frac{dx}{\left(x + \frac{3}{2}\right)^2 - \left(\frac{\sqrt{5}}{2}\right)^2} \\ &= \int \frac{dx}{\left(x + \frac{3}{2} - \frac{\sqrt{5}}{2}\right)\left(x + \frac{3}{2} + \frac{\sqrt{5}}{2}\right)} = \int \left(\frac{A}{x + \frac{3}{2} - \frac{\sqrt{5}}{2}} + \frac{B}{x + \frac{3}{2} + \frac{\sqrt{5}}{2}} \right) dx \\ &\quad A \left(x + \frac{3}{2} + \frac{\sqrt{5}}{2} \right) + B \left(x + \frac{3}{2} - \frac{\sqrt{5}}{2} \right) = 1 \end{aligned}$$

$$x(A + B) + A \left(\frac{3 + \sqrt{5}}{2} \right) + B \left(\frac{3 - \sqrt{5}}{2} \right) = 1$$

Equate the coefficients of like terms.

$$\left. \begin{array}{l} A + B = 0 \\ A \left(\frac{3 + \sqrt{5}}{2} \right) + B \left(\frac{3 - \sqrt{5}}{2} \right) = 1 \end{array} \right\} \implies A = \frac{1}{\sqrt{5}}, \quad B = -\frac{1}{\sqrt{5}}$$

Substitute the values into A and B .

$$\begin{aligned} I &= \int \left(\frac{A}{x + \frac{3}{2} - \frac{\sqrt{5}}{2}} + \frac{B}{x + \frac{3}{2} + \frac{\sqrt{5}}{2}} \right) dx \\ &= \int \left(\frac{1}{\sqrt{5} \left(x + \frac{3}{2} - \frac{\sqrt{5}}{2} \right)} - \frac{1}{\sqrt{5} \left(x + \frac{3}{2} + \frac{\sqrt{5}}{2} \right)} \right) dx \\ &= \boxed{\frac{1}{\sqrt{5}} \left(\ln \left| x + \frac{3}{2} - \frac{\sqrt{5}}{2} \right| - \ln \left| x + \frac{3}{2} + \frac{\sqrt{5}}{2} \right| \right) + C, \quad C \in \mathbb{R}} \end{aligned}$$

e. The integral:

$$I = \int \frac{\sin x}{1 + \sin x} dx = \int \frac{1 + \sin x - 1}{1 + \sin x} dx = \int dx - \int \frac{dx}{1 + \sin x}$$

The integral on the left evaluates to $x + C_1$. Evaluate the other integral.

$$\begin{aligned} \int \frac{dx}{1 + \sin x} &= \int \frac{1 - \sin x}{(1 + \sin x)(1 - \sin x)} dx = \int \frac{1 - \sin x}{1 - \sin^2 x} dx = \int \frac{1 - \sin x}{\cos^2 x} dx \\ &= \int (\sec^2 x - \tan x \sec x) dx = \tan x - \sec x + C_2 \end{aligned}$$

So, the result is

$$\boxed{I = x - \tan x + \sec x + C, \quad C \in \mathbb{R}}$$

5. If the function $y = f(x) \geq 0$ is continuously differentiable on $[a, b]$, the area of the surface generated by revolving the graph of $y = f(x)$ about the x -axis is

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

Find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = e^x$$

Set $a = 0$ and $b = 1$ and then evaluate the integral.

$$S = \int_0^1 2\pi e^x \sqrt{1 + (e^x)^2} dx$$

Let $u = e^x$, then $du = e^x dx$.

$$x = 0 \implies u = e^0 = 1, \quad x = 1 \implies u = e^1 = e$$

$$S = \int_0^1 2\pi e^x \sqrt{1 + (e^x)^2} dx = 2\pi \int_1^e \sqrt{1 + u^2} du$$

We will now use a trigonometric substitution. Let $u = \tan t$ for $0 < t < \frac{\pi}{2}$, then $du = \sec^2 t dt$.

$$\begin{aligned} S &= 2\pi \int_1^e \sqrt{1 + u^2} du = 2\pi \int \sqrt{1 + \tan^2 t} \cdot \sec^2 t dt = 2\pi \int \sqrt{\sec^2 t} \cdot \sec^2 t dt \\ &= 2\pi \int |\sec t| \sec^2 t dt = 2\pi \int \sec^3 t dt \quad [\sec t > 0] \end{aligned}$$

Find the antiderivative of $\sec^3 t$ with the help of integration by parts.

$$\begin{aligned} w &= \sec t \rightarrow dw = \sec t \tan t dt \\ dz &= \sec^2 t dt \rightarrow z = \tan t \end{aligned}$$

$$\begin{aligned} \int \sec^3 t du &= \tan t \cdot \sec t - \int \tan^2 t \sec t dt = \tan t \cdot \sec t - \int \frac{1 - \cos^2 t}{\cos^3 t} dt \\ &= \tan t \cdot \sec t - \int \sec^3 t dt + \int \sec t dt \end{aligned}$$

Notice that the integral appears on the right side of the equation. Therefore,

$$\int \sec^3 t dt = \frac{1}{2} \cdot \tan t \cdot \sec t + \frac{1}{2} \cdot \int \sec t dt$$

The integral of $\sec t$ with respect to t is

$$\int \sec t dt = \ln |\tan t + \sec t| + C_1, \quad c_1 \in \mathbb{R}$$

Recall $u = \tan t$.

$$u = \tan t \implies u^2 = \tan^2 t = \sec^2 t - 1 \implies \sec t = \sqrt{u^2 + 1}$$

$$\begin{aligned}
S &= 2\pi \cdot \frac{1}{2} (\tan t \cdot \sec t + \ln |\tan t + \sec t|) + C = \pi \left[u \cdot \sqrt{u^2 + 1} + \ln \left| t + \sqrt{u^2 + 1} \right| \right]_1^e \\
&= \pi \left[\left(e \cdot \sqrt{e^2 + 1} + \ln \left| e + \sqrt{e^2 + 1} \right| \right) - \left(\sqrt{2} + \ln \left| 1 + \sqrt{2} \right| \right) \right] \\
&= \pi \left[e \cdot \sqrt{e^2 + 1} - \sqrt{2} + \ln \left(\frac{e + \sqrt{e^2 + 1}}{1 + \sqrt{2}} \right) \right]
\end{aligned}$$

6. Let $u = 1 + e^{-x}$, then $du = -e^{-x} dx$. Handle the improper integral by taking the limit.

$$x = 0 \implies u = 1 + e^0 = 2, \quad x \rightarrow -\infty \implies u \rightarrow \infty$$

$$\begin{aligned}
\int_0^{-\infty} \frac{e^{-x}}{1 + e^{-x}} dx &= \int_2^{\infty} -\frac{du}{u} = \lim_{R \rightarrow \infty} \int_2^R -\frac{du}{u} = \lim_{R \rightarrow \infty} (-\ln u) \Big|_2^R \\
&= \lim_{R \rightarrow \infty} (-\ln R + \ln 2) = \boxed{-\infty}
\end{aligned}$$

The integral diverges to negative infinity.

7. According to the Monotone Convergence Theorem, if a sequence is both bounded and monotonic, the sequence converges. Take the corresponding function $f(x) = \frac{\ln x}{x}$. Apply the first derivative test and find the extrema.

$$f'(x) = \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$$

$$f'(x) = 0 \implies 1 - \ln x = 0 \implies \ln x = 1 \implies x = e$$

A critical point occurs at $x = e$. Apply the second derivative test and determine whether this is a local minimum or a local maximum.

$$f''(x) = \frac{-\frac{1}{x} \cdot x^2 - (1 - \ln x) \cdot 2x}{x^4} = \frac{2 \ln x - 3}{x^3}, \quad f''(e) = \frac{-1}{e^3} < 0$$

Therefore, this is a local maximum.

The first term of the sequence is $\frac{\ln 1}{1} = 0$. Take the limit at infinity. We may apply L'Hôpital's rule because we have an indeterminate form, which is ∞/∞ .

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} \stackrel{\text{L.H.}}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Since $\frac{\ln x}{x}$ is decreasing and bounded above by $f(e) = \frac{1}{e}$ and below by 0 for $x \geq e$, by the Monotone Convergence Theorem, the sequence converges.